

Q1. Genetic Data

(a) Differences between Gene Expression and SNP Data

Gene Expression Data measure the activity level of genes—how much mRNA is transcribed from each gene under certain conditions. It reflects dynamic cellular states, such as responses to stress, disease, or treatment [2].

Single Nucleotide Polymorphism (SNP) Data, on the other hand, represent static genetic variations—single-base changes in DNA sequences among individuals. SNPs are useful for studying genetic predispositions to diseases or population genetics, but they do not directly indicate gene activity levels [5].

Feature	Gene Expression	SNP
Type	mRNA (functional output)	DNA (genomic variant)
Dynamic/Static	Dynamic (changes by condition)	Static (inherited)
Purpose	Functional genomics	Genomic variation
Example Use	Cancer subtype classification	GWAS (Genome-Wide Association Study) [5]

(b) Realizations of Next-Generation Sequencing (NGS)

One major realization of NGS is **RNA-Seq** (RNA Sequencing), which quantifies gene expression using high-throughput sequencing [4]. It has become a revolutionary tool for studying the transcriptome [1], [2].

Workflow Illustration [1], [3]:

1. **RNA Extraction** – mRNA is isolated from the biological sample.
2. **cDNA Synthesis** – mRNA is reverse-transcribed into complementary DNA (cDNA).
3. **Library Preparation** – cDNA fragments are ligated with sequencing adapters.
4. **Sequencing** – Libraries are sequenced using platforms like Illumina, producing millions of short reads.
5. **Data Analysis** – Reads are aligned to a reference genome and quantified to estimate expression levels (in TPM/FPKM units) [1].

(c) Improvements of NGS over Microarray

Key Improvement: Higher Sensitivity and Dynamic Range [2], [4]

Microarrays rely on pre-designed probes, limiting detection to known genes and suffering from cross-hybridization noise. NGS, however, directly sequences transcript molecules, enabling:

- Detection of low-abundance transcripts that microarrays miss.
- Unbiased discovery of novel isoforms or non-coding RNAs.
- Quantitative accuracy across a broader range (over 10^5 -fold difference in expression levels).

This improvement allows researchers to capture the full complexity of the transcriptome, revealing subtle regulatory changes that microarrays often obscure [2].

Q2. Dynamic Interaction Network and Relevance Value

(a) Matrix Elements

The equation can be expressed in the matrix form for a single patient .

Assuming a set of genes, the elements are:

- $X(n)$ (Gene Expression Vector): A $p \times 1$ column vector representing the expression levels of all p genes for patient n .
- A (Interaction Matrix): A $p \times p$ matrix where the element a_{ij} represents the interaction ability of gene j on gene i . A special property, derived

from the $j=i$ condition, is that all diagonal elements of A are zero ($a_{ii}=0$).

- **(Error Vector):** A column vector representing the stochastic noise for each gene for patient.

(b) Solving for A^+ and A^-

The interaction matrices were solved by treating each row of the matrix as an independent linear regression problem. For each gene, its expression level (y) was modeled as a linear combination of the expression levels of all other genes (X). This was solved using Ridge Regression with an L2-regularization coefficient (α) of 1, as specified. The `fit_intercept` parameter was set to `False` as the model does not include a constant term.

```
def solve_interaction_matrix(X_group):  
    """  
    Solves the gene interaction matrix A for a given patient group  
    using row-by-row Ridge Regression.  
  
    Args:  
        X_group (np.ndarray): Gene expression data for a group,  
                               shape (n_patients, p_genes).  
  
    Returns:  
        np.ndarray: The solved interaction matrix A, shape (p_genes, p_genes).  
    """  
    # Get the number of genes, p  
    _, p_genes = X_group.shape  
  
    # Initialize an empty p x p matrix for A  
    A = np.zeros((p_genes, p_genes))  
  
    # Create a Ridge Regression model with alpha=1 and no intercept,  
    # as the model  $x_i = \sum(a_{ij} * x_j)$  has no constant term.  
    model = Ridge(alpha=1, fit_intercept=False)  
  
    # Solve for each row of A by treating each gene as a target  
    for i in range(p_genes):  
        # The target variable (y) is the expression of the i-th gene  
        y_target = X_group[:, i]  
  
        # The features (X) are the expressions of all other genes ( $j \neq i$ ).  
        # This is done by deleting the i-th column from the data matrix.  
        X_features = np.delete(X_group, i, axis=1)  
  
        # Fit the model using the features and target  
        model.fit(X_features, y_target)  
  
        # Get the learned coefficients, which correspond to the i-th row of A  
        coeffs = model.coef_  
  
        # Insert the coefficients into the i-th row of the final A matrix,  
        # placing a 0 at the diagonal position ( $a_{ii} = 0$ ).  
        A[i, :] = np.insert(coeffs, i, 0)  
  
    return A
```

(c) Relevance Value Calculation

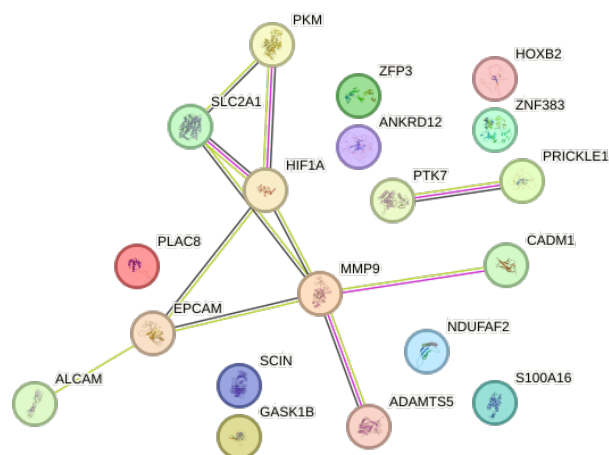
The Relevance Value (RV) for each gene quantifies the total change in its interaction influence between the positive and negative groups. It was calculated by taking the absolute difference between the interaction matrices and summing the values down each column. The top 15 genes with the highest RVs are listed below.

Rank	Gene	Relevance Value
1	PLAC8	4.9781
2	MMP9	4.8135
3	FAM198B	4.5533
4	PRICKLE1	4.0398
5	ZFP3	4.0019
6	SNORD89	3.9309
7	LOC202181	3.9125
8	ZNF383	3.8391
9	LOC389765	3.8162
10	S100A16	3.7446
11	NDUFAF2	3.6928
12	SCIN	3.5561
13	ANKRD12	3.5531
14	HOXB2	3.4953
15	ADAMTS5	3.4848

Q3. Biological Enrichment

(a) Interaction Graph of Top-15 RV Genes and Biomarkers

The following network graph from the STRING database shows the known and predicted interactions between the top-15 RV genes and the seven provided biomarkers.



(b) Biological Functions of the RV-based Network

Enrichment analysis of this network revealed several key functions related to cancer progression.

1. Central Carbon Metabolism in Cancer (KEGG: hsa05230)

This pathway describes the "Warburg effect," a metabolic shift where cancer cells favor rapid glucose fermentation to fuel growth. Key players like **PKM** and **SLC2A1** in our network are central to this process, which allows for the rapid production of cellular building blocks.

2. Hypoxia-Inducible Factor 1 (HIF-1) Signaling Pathway

This pathway, driven by **HIF1A**, is a cell's survival response to low-oxygen (hypoxic) conditions found in solid tumors. Activating this pathway promotes new blood vessel growth (angiogenesis) and metastasis, key drivers of tumor aggressiveness.

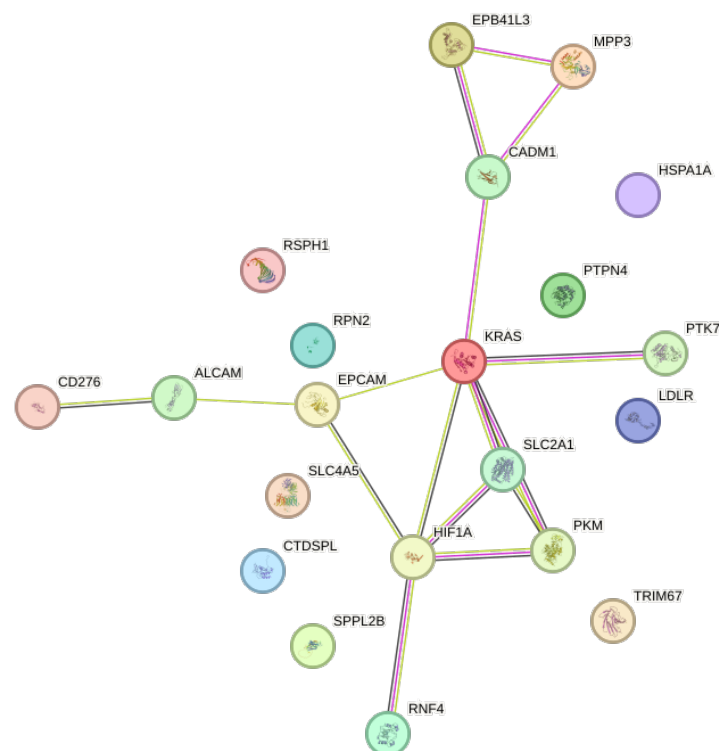
3. Cell Adhesion and Extracellular Matrix (ECM) Remodeling

This function is crucial for metastasis. It involves the disruption of cell-cell connections (mediated by **CADM1**, **ALCAM**, **EPCAM**) and the degradation of the surrounding tissue scaffold by enzymes like **MMP9** and **ADAMTS5**, allowing cancer cells to invade and spread.

(c) Analysis of CADM1 Physical Interactors

A second analysis was performed using genes known to physically interact with **CADM1** (sourced from the BioGrid database).

Interaction Graph of CADM1 Interactors



Biological Functions of the CADM1 Interactor Network

This focused network highlighted more specific, mechanistic pathways.

1. Central Carbon Metabolism in Cancer (KEGG: hsa05230)

The enrichment of this pathway suggests that **CADM1**-associated scaffolding proteins (e.g., **DLG1**, **SCRIB**) may organize signaling complexes that influence

growth factor pathways, ultimately driving the metabolic switch to support tumorigenesis.

2. Renal Cell Carcinoma (KEGG: hsa05211)

The enrichment of this specific cancer type is a powerful finding. **CADM1** and its interacting partners are involved in maintaining cell polarity and junction organization. The dysfunction of these tumor-suppressing proteins can lead to the loss of tissue architecture, a key step in the development of kidney cancer.

3. Thyroid Hormone Signaling Pathway (KEGG: hsa04919)

This pathway's enrichment suggests crosstalk between cell adhesion and growth signaling. **CADM1** and its partners are located at the cell membrane where growth factor receptors reside. Disruption of these complexes can alter the cell's sensitivity to growth signals, creating a pro-tumorigenic environment.

References

- [1] A. Mortazavi, B. A. Williams, K. McCue, L. Schaeffer, and B. Wold, "Mapping and quantifying mammalian transcriptomes by RNA-Seq," *Nature Methods*, vol. 5, no. 7, pp. 621–628, 2008.
- [2] Z. Wang, M. Gerstein, and M. Snyder, "RNA-Seq: a revolutionary tool for transcriptomics," *Nature Reviews Genetics*, vol. 10, no. 1, pp. 57–63, 2009.
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- [4] M. L. Metzker, "Sequencing technologies — the next generation," *Nature Reviews Genetics*, vol. 11, no. 1, pp. 31–46, 2010.
- [5] A. K. Pabinger, et al., "A survey of tools for variant analysis of next-generation genome sequencing data," *Briefings in Bioinformatics*, vol. 15, no. 2, pp. 256–278, 2014.